Title: Industrial Alarm Logic.

Objectives:

- To implement alarm logic instruction in Rockwell CCW.

System Description:

In industrial environments, alarm systems do not simply turn on and off with a sensor, they require human interaction to ensure operators are aware of faults. This system simulates an alarm management process.

The operator starts the main system (M101_ONCMD). An alarm condition occurs (simulated by the HOOTER signal). An Alarm Indicator Lamp (ACK_LAMP) illuminates. The operator must press an Acknowledge pushbutton (ACK_PB) to confirm they have seen the alarm. This latches the lamp ON. Once the fault is cleared, the operator presses a Reset pushbutton (RESET) to turn the indicator lamp OFF and reset the alarm logic.



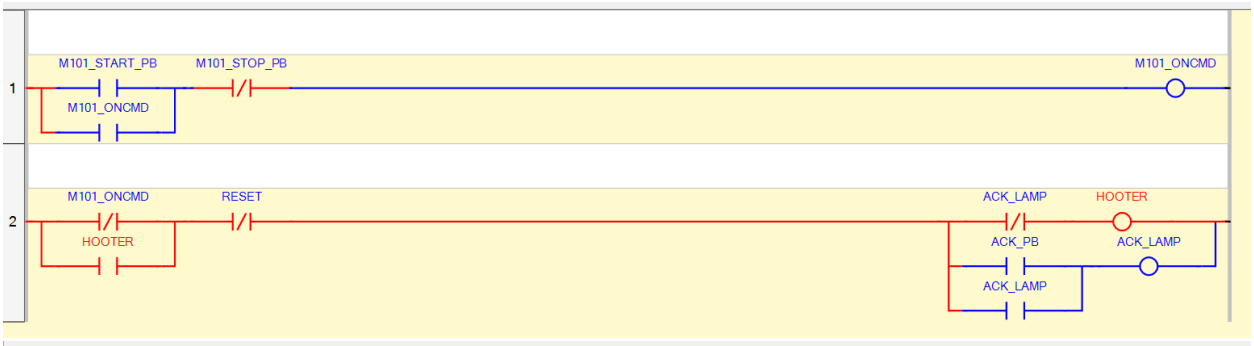
The Variables of the systems are:

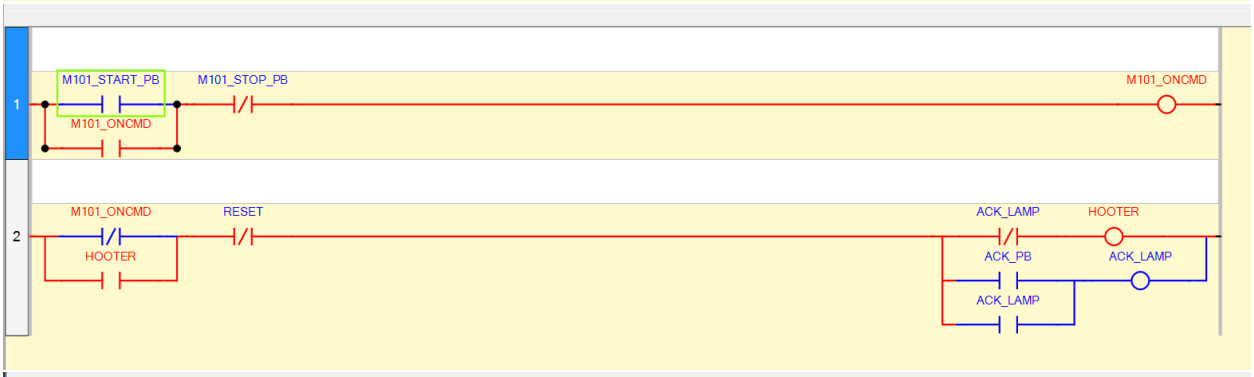
	Name	Alias	Data Type	Dimension	Initial Value	Comment	String Size
			BO				
▶	M101_START_PB		BOOL				
	M101_ONCMD		BOOL				
	M101_STOP_PB		BOOL				
	HOOTER		BOOL				
	ACK_PB		BOOL				
	ACK_LAMP		BOOL				
	RESET		BOOL				
*							

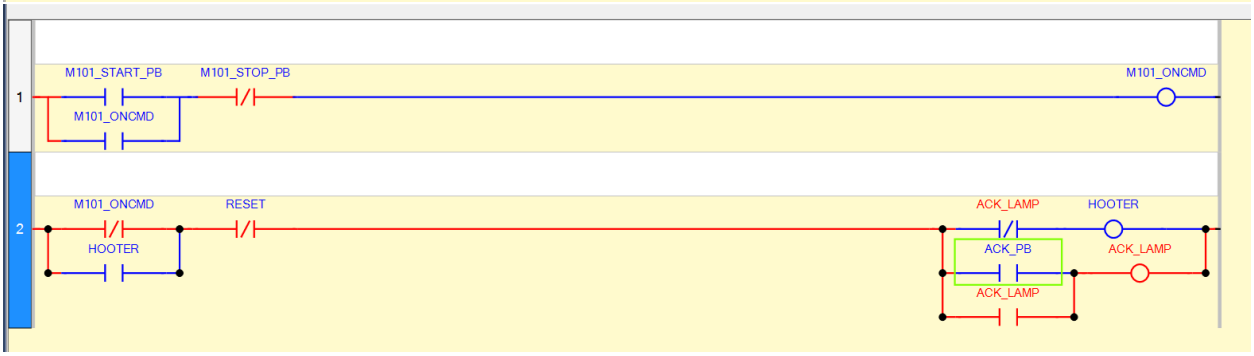
The input/ouput (I/O) mapping is:

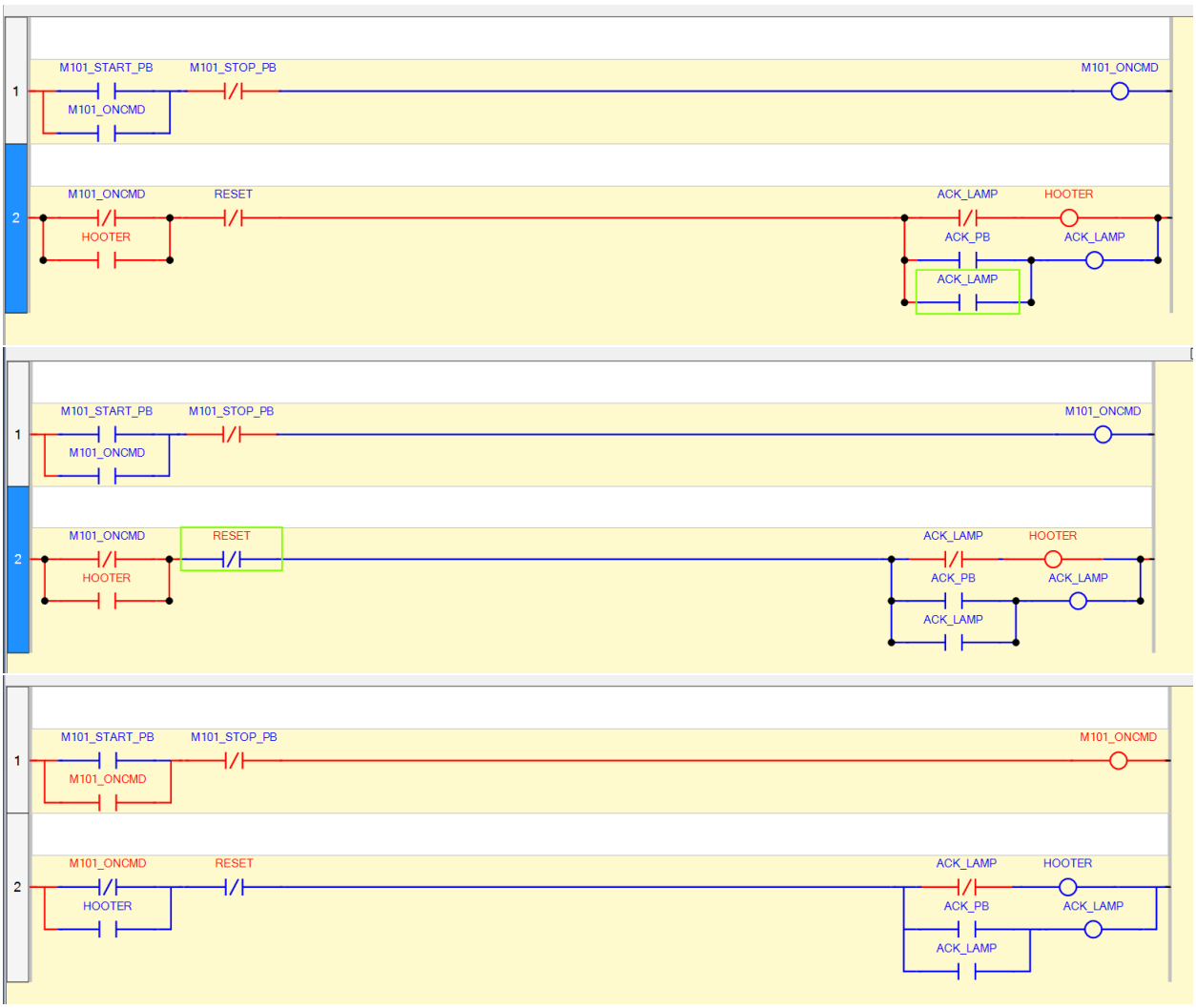

1
_IO_EM_DI_01
M101_START_PB
2
_IO_EM_DI_02
M101_STOP_PB
3
_IO_EM_DI_03
ACK_PB
4
_IO_EM_DI_04
RESET
5
M101_ONCMD
_IO_EM_DO_03
6
HOOTER
_IO_EM_DO_02
7
ACK_LAMP
_IO_EM_DO_01

Output:


1
M101_START_PB
M101_STOP_PB
M101_ONCMD
M101_ONCMD
2
M101_ONCMD
RESET
ACK_LAMP
HOOTER
HOOTER
ACK_PB
ACK_LAMP
ACK_LAMP


1
M101_START_PB
M101_STOP_PB
M101_ONCMD
M101_ONCMD
2
M101_ONCMD
RESET
ACK_LAMP
HOOTER
HOOTER
ACK_PB
ACK_LAMP
ACK_LAMP


1
M101_START_PB
M101_STOP_PB
M101_ONCMD
M101_ONCMD
2
M101_ONCMD
RESET
ACK_LAMP
HOOTER
HOOTER
ACK_PB
ACK_LAMP
ACK_LAMP



Discussion:

We first press the “START” push button to turn on the system (“M101_ONCMD”). When an alarm condition occurs, the “HOOTER” input activates and turns on the “ACK_LAMP”. We then press the “ACK_PB” push button to acknowledge the alarm, this latches the lamp ON so it stays on even if the initial alarm clears. To clear the alarm after fixing the issue, we press the “RESET” push button, which breaks the circuit and turns off the “ACK_LAMP”. The whole system can be turned off at any time using the “STOP” push button.

Conclusion:

This lab successfully demonstrated the design and simulation of an industrial alarm management system with acknowledge and reset functions using Rockwell CCW. The experiment reinforced the practical application of fundamental PLC programming concepts, specifically I/O hardware mapping techniques and complex latching circuits for operator safety interfaces.